LECTURE TWO • 2026

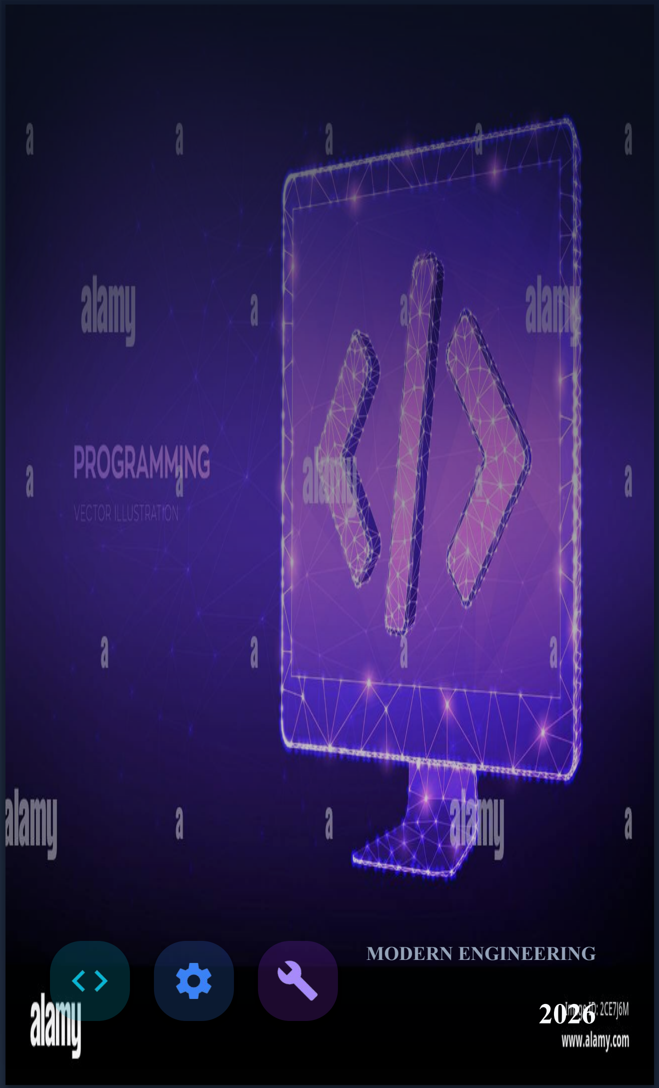
Software Engineering

Understanding Software Fundamentals & Engineering Principles

Software Development

Process Models

Quality Assurance



What is Software?

A textbook description of software consists of three key components:

1 **Instructions**

Computer programs that provide desired function and performance

2 **Data Structures**

Enable programs to adequately manipulate information

3 **Documents**

Describe the operation and use of the programs

Software Characteristics

Software has characteristics considerably different from hardware:

Logical, Not Physical

A logical system element

Engineered

Developed, not manufactured

Doesn't "Wear Out"

No spare parts needed

Custom Built

Most software is custom built

> Every software failure indicates an **error in design** or in the process through which design was translated into machine executable code.

COMPARISON

Hardware vs Software

Software Applications — Part 1

Software areas indicate the breadth of potential applications:

System Software

Programs that service other programs (compilers, editors, file management utilities)

Real-Time Software

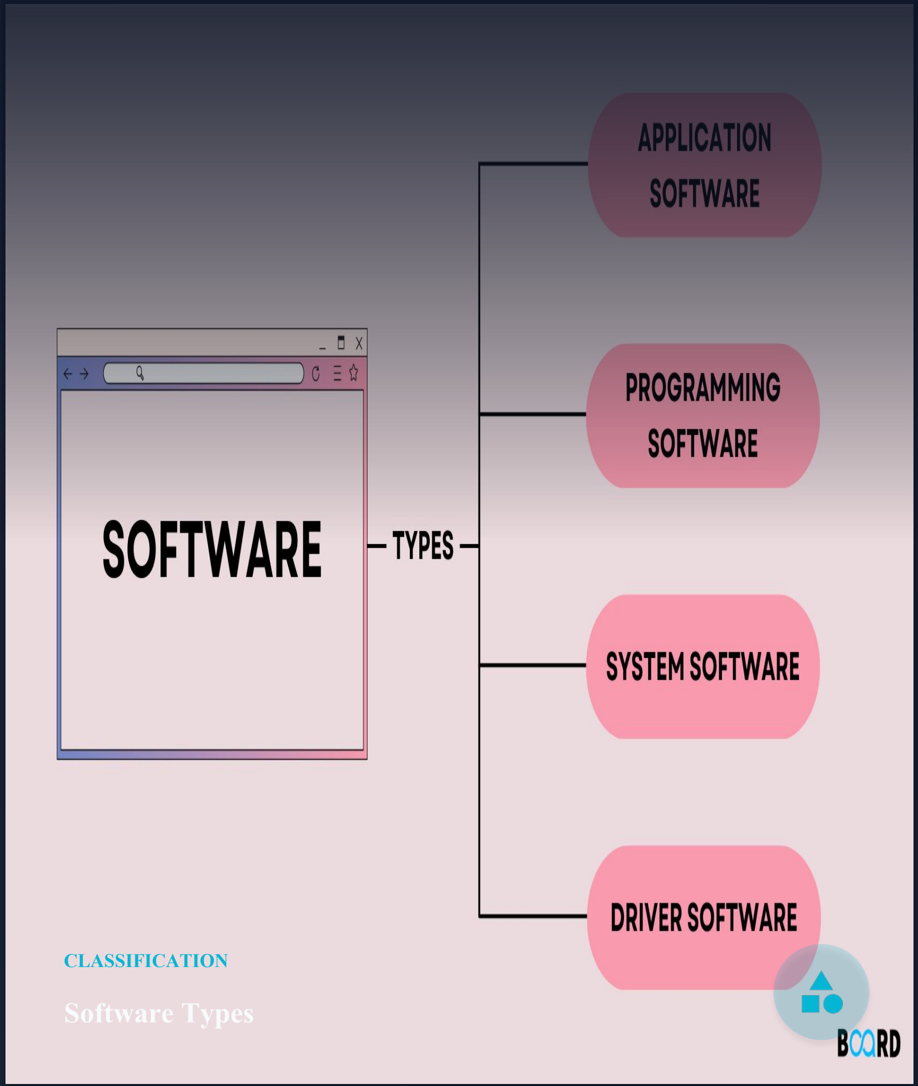
Monitors/analyzes/controls real-world events as they occur

Business Software

Largest single software application area for information processing

Engineering & Scientific

Characterized by "number crunching" algorithms and computations



CLASSIFICATION

Software Types

Software Applications — Part 2

Embedded Software

Intelligent products in consumer and industrial markets

PC Software

Word processing, spreadsheets, graphics, multimedia

Web-Based Software

Web pages with executable instructions (CGI, HTML, Java)

AI Software

Non-numerical algorithms for complex problem solving

The personal computer software market has burgeoned over the past two decades with hundreds of applications across **entertainment, business, and networking**.

AI Software

Definitions of Software Engineering

01

The **establishment and use of sound engineering principles** in order to obtain economically software that is **reliable and works efficiently** on real machines.

02

The **application of a systematic, disciplined, quantifiable approach** to the **development, operation, and maintenance** of software; that is, the application of engineering to software.

Systematic

Quantifiable

Engineering Principles

Software Engineering Layers

Software engineering is a **layered technology** with three essential components:

Process Layer

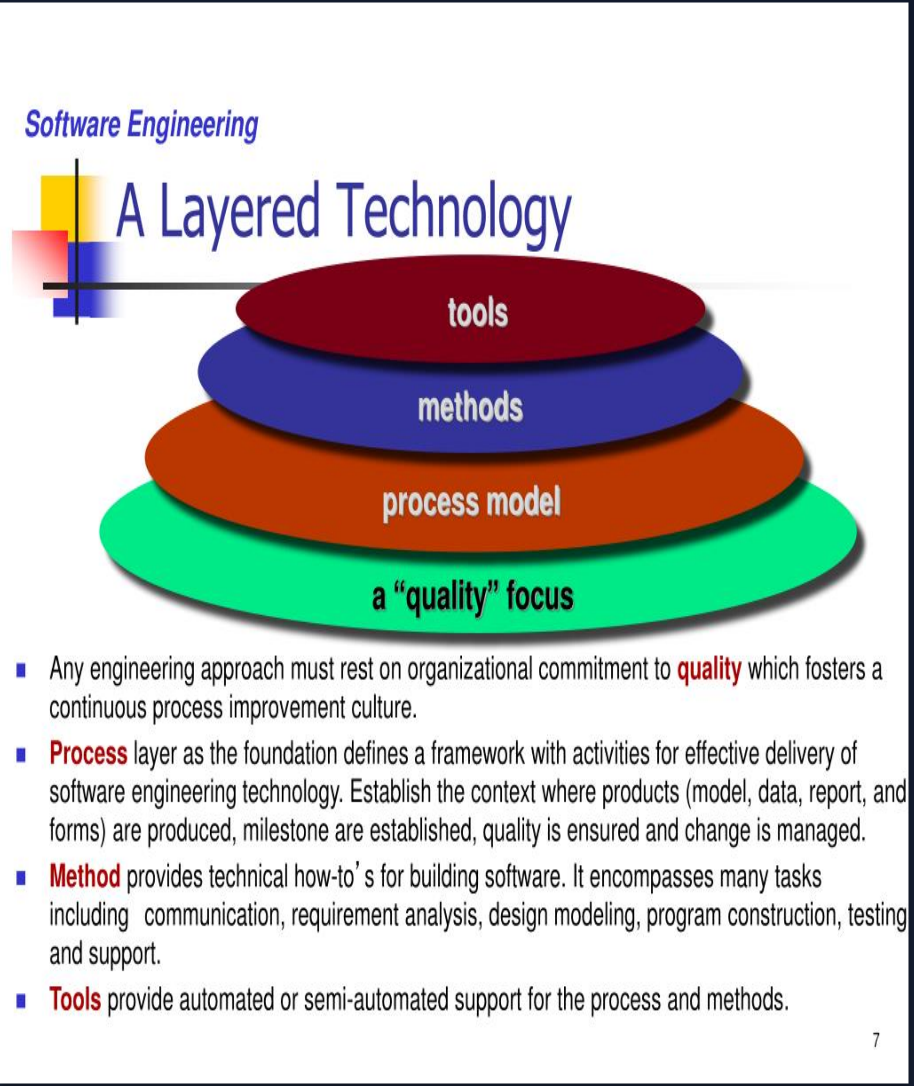
The **foundation** — the glue that holds technology layers together and enables rational and timely development

Methods

Provide **technical how-to's** — requirements analysis, design, construction, testing, and support Software engineering methods rely on a set of basic principles that govern each area of the technology and include modelling activities and other descriptive techniques.

Tools

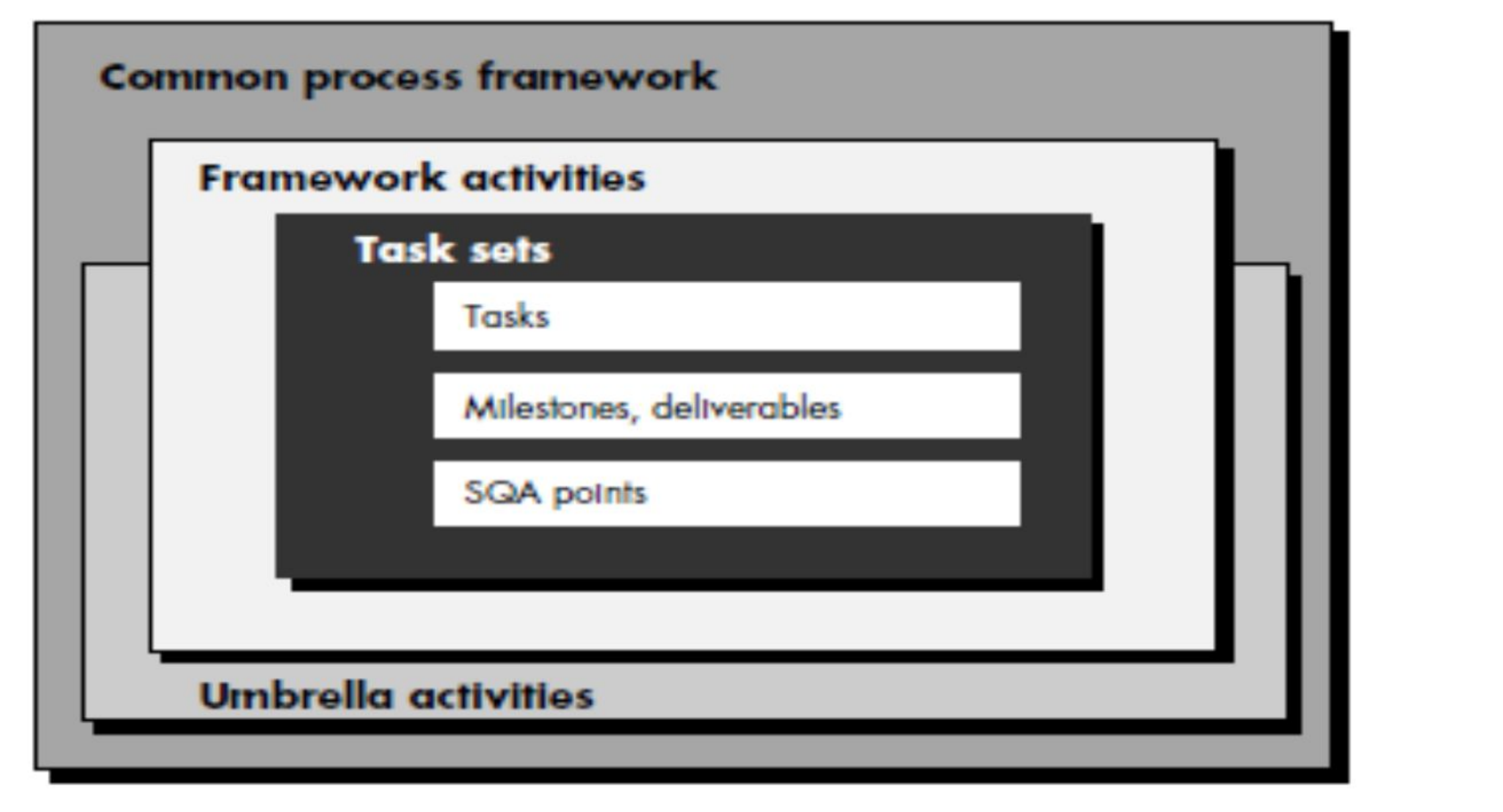
Provide **automated support** — integrated tools form computer-aided software engineering When tools are integrated so that information created by one tool can be used by another, a system for the support of software development, called computer-aided software engineering, is established.



ARCHITECTURE

Engineering Layers

The software process



A **common process framework** is established by defining a small number of **framework activities**

The Software Process

Framework Activities

Applicable to **all software projects** regardless of size or complexity

Task Sets

Enable **adaptation** to project characteristics through work tasks, milestones, and quality assurance points

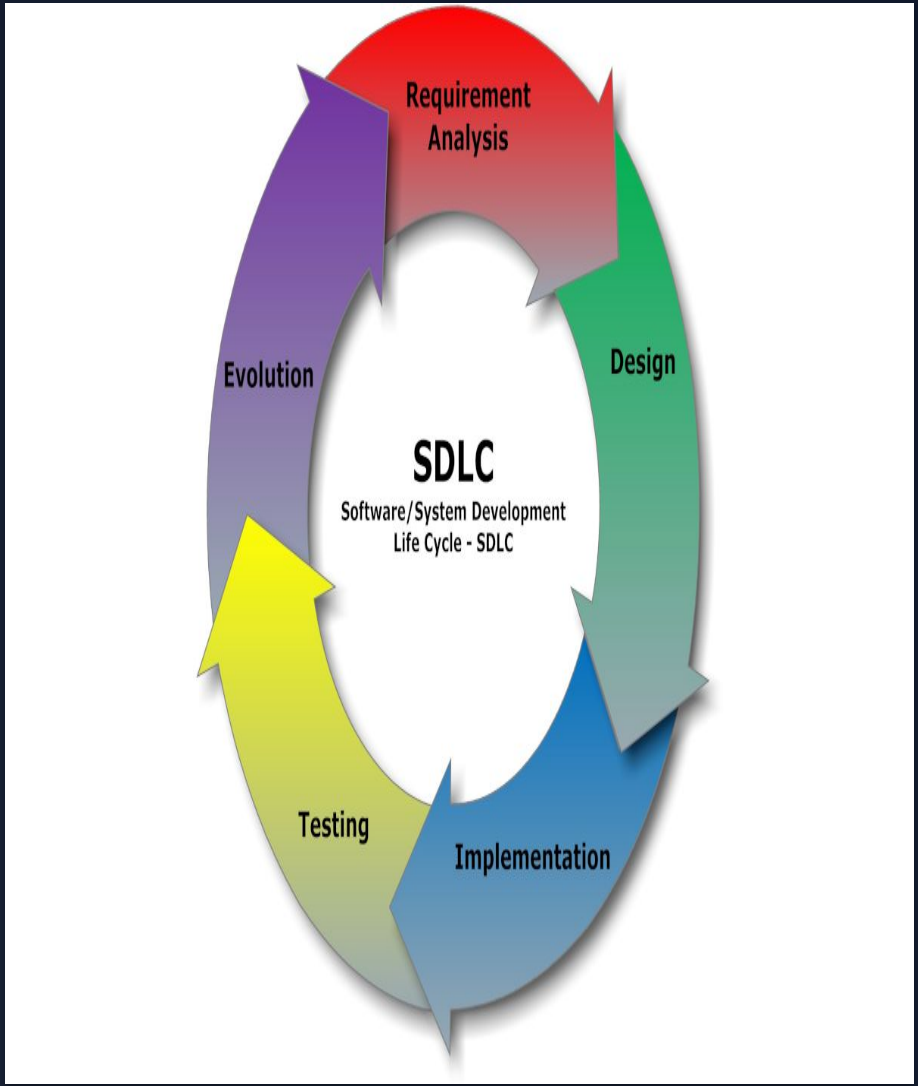
Umbrella Activities

Overlay the process model — quality assurance, configuration management, and measurement occur throughout the process

Quality Assurance

Configuration

Measurement



PROCESS MODEL

Software Lifecycle